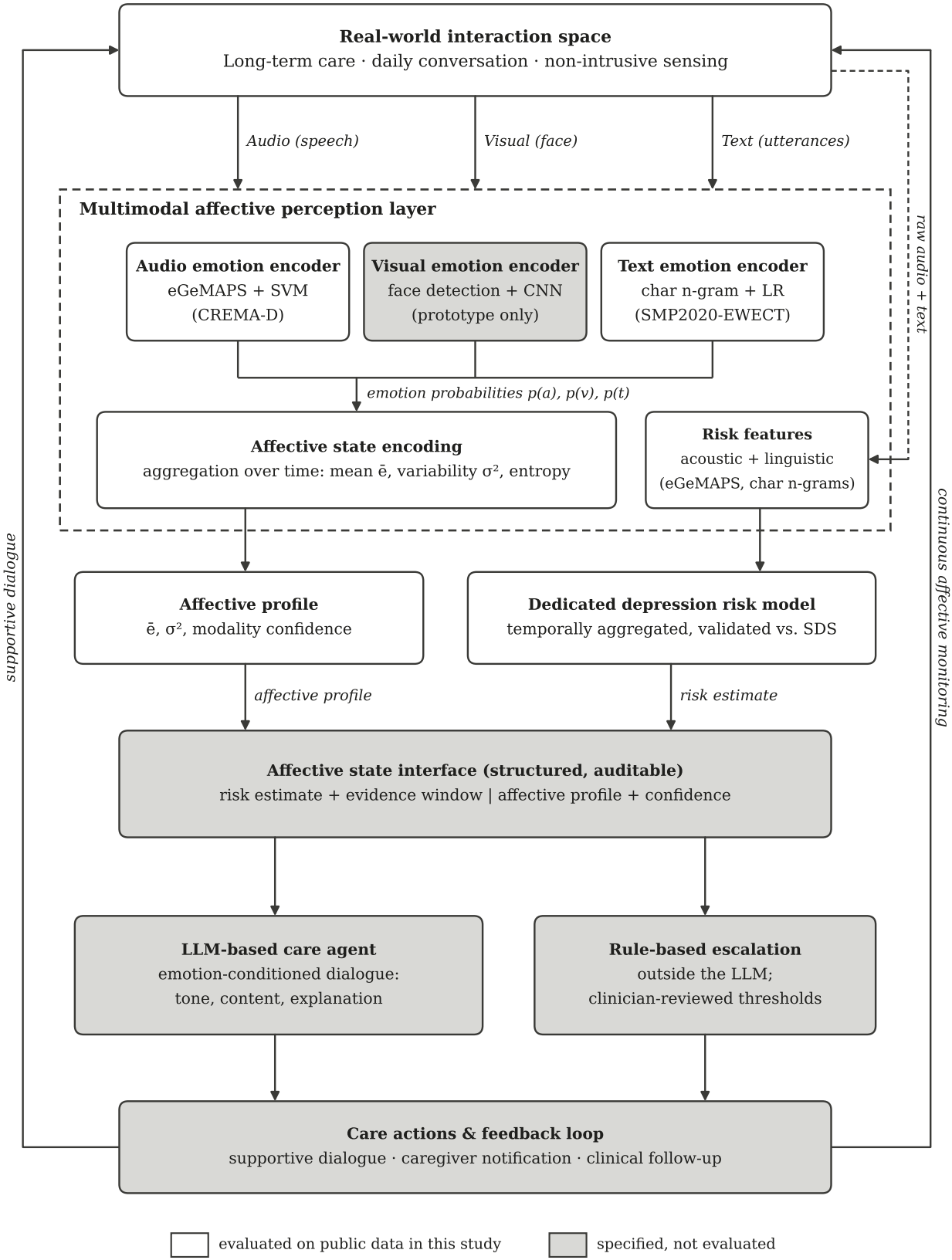
Real-world interaction space
Long-term care · daily conversation · non-intrusive sensing
Audio (speech)
Visual (face)
Text (utterances)
Multimodal affective perception layer
Audio emotion encoder
eGeMAPS + SVM
(CREMA-D)
Visual emotion encoder
face detection + CNN
(prototype only)
Text emotion encoder
char n-gram + LR
(SMP2020-EWECT)
raw audio + text
emotion probabilities p(a), p(v), p(t)
Affective state encoding
aggregation over time: mean ē, variability σ², entropy
Risk features
acoustic + linguistic
(eGeMAPS, char n-grams)
supportive dialogue
continuous affective monitoring
Affective profile
ē, σ², modality confidence
Dedicated depression risk model
temporally aggregated, validated vs. SDS
affective profile
risk estimate
Affective state interface (structured, auditable)
risk estimate + evidence window | affective profile + confidence
LLM-based care agent
emotion-conditioned dialogue:
tone, content, explanation
Rule-based escalation
outside the LLM;
clinician-reviewed thresholds
Care actions & feedback loop
supportive dialogue · caregiver notification · clinical follow-up
evaluated on public data in this study
specified, not evaluated